

Gautham Raju

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Education

University of California, Berkeley

Master of Public Policy (MPP)

Berkeley, CA

Aug. 2026 – Present

The University of Texas at Austin

Bachelor of Science in Computer Science; GPA: 3.8/4.0

Austin, TX

Aug. 2019 – May 2023

Experience

Toyota Connected

Software Engineer II

Plano, TX

May 2025 – Aug 2026

- Built a production vehicle-information agent on Amazon Bedrock AgentCore as part of a five-person team, using Python and the Strands Agents SDK to connect an LLM to live data and image services through dynamically discovered Model Context Protocol (MCP) tools.
- Engineered the agent runtime with OAuth 2.0 machine-to-machine authentication, short-term conversation memory, retry and timeout handling, and cross-instance Redis/Valkey caching; compressed 100–300 KB tool responses and enabled on-demand detail retrieval to preserve LLM context-window capacity.
- Implemented defense-in-depth AI guardrails through pre- and post-tool hooks that enforce legal-disclaimer retrieval, brand scope, and required calls to action; added periodic compliance reminders and stream-time validation designed to prevent hallucinated links and exposure of internal disclaimer codes.
- Developed a configurable evaluation framework for a customer-facing RAG application with scheduled production sanity checks and manual runs across environments; generated synthetic test cases with templates and LLMs with tagged results.
- Implemented asynchronous LLM-as-judge and deterministic metrics for groundedness, query relevance, completeness, contextual precision and recall, semantic similarity, ground-truth accuracy, retrieval diversity, and disclaimer citation; calibrated thresholds and multi-model aggregation to test legal and organizational guardrails across alternate query phrasings and user personas.
- Operationalized evaluation and agent monitoring through Datadog dashboards, OpenTelemetry tracing, structured conversation logging via AWS SQS, Lambda, and MongoDB, and recurring, programmatically-generated HTML reports for technical and business stakeholders.

Toyota Connected

Software Engineer I

Plano, TX

Jan. 2023 – May 2025

- Automated California Consumer Privacy Act (CCPA) Right to Correct and Right to Delete workflows by designing and implementing a Go service that consumes AWS SQS requests and updates or deletes personal information in MongoDB and Salesforce.
- Researched, implemented, and deployed a STOMP-over-WebSocket service that routes Twilio live-call transcriptions to subscribed Salesforce agent sessions; secured message integrity and authenticity with AWS Key Management Service and reduced average call duration by more than 20 seconds.
- Prototyped a mobile public-safety service with two other engineers, building Java/Spring back-end and Salesforce agent workflows that share real-time location data and support escalation to public safety responders.

Public-Interest Project

MySky

Berkeley Center for Human-Compatible Artificial Intelligence (CHAI); PI: Jonathan Stray

June 2026 – Present

- Built the TypeScript, Lit, and MobX frontend for an open-source ATProto/Bluesky recommender designed to make prosocial feed ranking transparent and user-controllable; created responsive experiences for feed exploration, per-post “Why Am I Seeing This?” explanations, pipeline visualization, and feed-specific settings.
- Designed controls for recommendation-source weights, freshness, and feed purpose, including feed-scoped state, optimistic updates with isolated rollback, live previews of reranked content, pagination, and undo/redo behavior so users can inspect the effects of algorithmic choices before saving them.
- Extended the Python/FastAPI backend with Firebase-authenticated transparency and preference APIs, Bluesky post hydration, and Firestore pipeline snapshots; made candidate time windows, source weights, and purpose configurable per feed and persisted concurrent preference changes through transactional updates.
- Implemented the Bluesky OAuth bridge with Firebase Cloud Functions, PostHog analytics and feedback surveys, CI/CD workflows, and automated unit and end-to-end coverage with Vitest and Playwright.

Skills

AI Governance: BlueDot Impact Frontier AI Governance
Policy Analysis and Quantitative Methods: Policy analysis, policy memo writing, quantitative research, statistical inference, decision analysis, forecasting, stakeholder and institutional analysis, microeconomics
AI Engineering and Evaluation: Agentic AI, recommender systems, RAG, MCP, LLM-as-judge, synthetic test generation, multi-model evaluation, guardrail validation
Programming: Python, TypeScript, Java, Go, JavaScript, SQL, C++, C#
Platforms and Frameworks: Amazon Bedrock, AgentCore, Strands Agents SDK, AWS SQS/Lambda/KMS, Git, Datadog, Docker, Kubernetes, MongoDB, TensorFlow, OAuth 2.0

Community Leadership

Bold Idea: Coding club mentor in underserved communities; technology sustainability committee member
ACEing Autism: Volunteer tennis buddy